Injective PEPS Fundamental Theorem: Section 3 Route

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1 The source argument

The injective PEPS part of [MGRPG+18] proves the Fundamental Theorem for tensor networks on finite graphs without double edges and without self-loops. The proof is local. It does not slice a two-dimensional network into a long one-dimensional chain. Instead, it fixes an edge, blocks the graph around that edge into a three-site injective MPS, applies the one-dimensional injective theorem, and repeats this construction over all edges.[1]

For a chosen edge $e = (u, v)$, all vertices other than u and v are grouped into the middle tensor. The two endpoint tensors and the middle tensor form a three-site MPS. Injectivity of the PEPS tensors implies injectivity of this blocked MPS. Since the two original PEPS represent the same state, the two blocked MPS represent the same state. The one-dimensional injective theorem therefore gives an algebra isomorphism between matrix insertions on the chosen bond. Equivalently, an arbitrary matrix X inserted on that bond in one network is matched by a matrix on the corresponding bond in the other network.

After choosing the resulting edge gauges for every edge and absorbing them into one tensor family, the paper obtains the following stronger comparison: for every edge and every matrix X, inserting X on that edge in the first PEPS gives the same state as inserting X on the same edge in the modified second PEPS.[2]

2 Current formal representation

The finite graph, edge, incident-edge, tensor, virtual configuration, state coefficient, equality of states, and vertex injectivity are represented in `TNLean/PEPS/Defs.lean`. Vertex injectivity is the linear independence of the local physical vectors indexed by local virtual configurations, matching the paper's statement

[1] The edge-blocking construction is `Papers/1804.04964/paper_normal.tex:981--1009`. The equality of the two PEPS before blocking is at `Papers/1804.04964/paper_normal.tex:1010--1036`. The application of the three-site injective MPS theorem is at `Papers/1804.04964/paper_normal.tex:1037--1065`.

[2] This is the displayed equation labelled `eq:inj_equal_edge` in `Papers/1804.04964/paper_normal.tex:1037--1065`.

that each tensor is injective as a map from the virtual space to the physical space.

The local linear-algebraic realization of virtual operations at an injective vertex is represented in `TNLean/PEPS/VirtualInsertion.lean`. The chosen left inverse of the local tensor map and the resulting physical realization of local virtual endomorphisms correspond to the same observation used in the three-site MPS proof: a virtual operation can be represented by a physical operation on a neighboring site. The basis computation TNLean.PEPS.localIncidentMatrixOp˙single and its tensor-map form TNLean.PEPS.localTensorMap˙localIncidentMatrixOp˙single identify the one-edge matrix action on a single local virtual boundary with the expected sum over the new distinguished edge index.

The edge-centered blocking data are represented in `TNLean/PEPS/Blocking.lean`. The existing declarations split local virtual configurations at a distinguished incident edge and split the vertex product into the left endpoint, the middle region, and the right endpoint. The edge-blocked coefficient is proved equal to the original PEPS coefficient.

The next formal layer is the matrix-insertion comparison. The declarations TNLean.PEPS.EdgeInsertedBoundaryConfig, TNLean.PEPS.edgeBoundaryToInsertedBoundaryConfig, TNLean.PEPS.EdgeDiagonalInsertedBoundaryConfig, TNLean.PEPS.edgeBoundaryConfigEquivDiagonalInsertedBo TNLean.PEPS.edgeMiddleConfigToOpenMiddleConfig, TNLean.PEPS.edgeOpenMiddleConfigToMiddleConfig, TNLean.PEPS.edgeMiddleConfigEquivOpenMiddleConfig, TNLean.PEPS.edgeOpenMiddleWeight, TNLean.PEPS.edgeBoundaryLeftIndexEquivInsertedBoundaryConfig, TNLean.PEPS.edgeBoundaryRightIndexEquiv and TNLean.PEPS.edgeInsertedCoeff represent the open edge contraction used when an arbitrary matrix is inserted on the distinguished edge. These definitions are the formal counterpart of the matrix insertion appearing before and after the equation labelled `eq:inj_equal_edge`. The two maps TNLean.PEPS.edgeMiddleConfigToOpenMiddleConfig and TNLean.PEPS.edgeOpenMiddleConfigToMiddleConfig have been packaged as the finite equivalence TNLean.PEPS.edgeMiddleConfigEquivOpenMiddleConfig. This isolates the reindexing needed for the identity-insertion specialization. The current Lean layer includes the corresponding middle-tensor reindexing theorem TNLean.PEPS.edgeMiddleWeight˙eq˙edgeOpenMiddleWeight. The diagonal summand has also been isolated in TNLean.PEPS.edgeInsertedCoeff˙identity˙diagonal˙summand, and TNLean.PEPS.edgeInsertedCoeff˙identity˙offDiagonal˙summand records the pointwise off-diagonal vanishing. The full identity-insertion specialization is TNLean.PEPS.edgeInsertedCoeff˙identity; its proof uses TNLean.PEPS.edgeBoundaryConfigEquivDiagonalInserted to reindex the finite inserted-boundary sum along the diagonal.

3 Remaining mathematical obligations

The following statements are still needed before the injective PEPS Fundamental Theorem can be proved in the manner of the paper.[3]

[3]This list and the ledger below are the status record referenced by the Section 3 docstrings and blueprint comments.

1. Vertex injectivity must be shown to pass to the edge-blocked three-site object. This is the formal version of the paper's assertion that the regrouped MPS is injective. The middle physical index for this regrouped object is now explicit as TNLean.PEPS.EdgeMiddlePhysicalConfig, and the middle contractions have the middle-indexed forms TNLean.PEPS.edgeMiddleWeightOn and TNLean.PEPS.edgeOpenMiddleWeightOn. In the blueprint the injectivity statement is TNLean.PEPS.EdgeBlockedThreeSiteInjective. The singleton base case is now formalized by TNLean.PEPS.singletonRegionTensorFamily and TNLean.PEPS.IsVertexInjective.singletonRegionTensorInjective: for a one-vertex block $\{v\}$, $T_{A,\{v\}}(\eta,\sigma) = A_v(\eta,\sigma)$, hence vertex injectivity gives $\mathrm{inj}(T_{A,\{v\}})$. The bridge to the abstract region-injectivity predicate is recorded by TNLean.PEPS.SingletonRegionInjectivityComparison and TNLean.PEPS.SingletonRegionInjectivityFromVertexInjectivity.ofSingletonComparison. With the abstract union-closure lemma, this gives every nonempty finite region by TNLean.PEPS.RegionInjectivityUnionClosure.finset˙injective˙of˙singletons: $R = \bigcup_{v\in R}\{v\}$ and $\kappa(\{v\})$ for every $v \in R$ imply $\kappa(R)$. The corresponding nonempty-middle edge-middle tensor consequence is recorded in `thm:peps_edgeMiddleTensorInjective_of_singletons`: for $M_e = V \setminus \{u,v\}$, the assumptions $M_e \neq \varnothing$ and $M_e = \bigcup_{w\in M_e}\{w\}$ give $\kappa(M_e)$; the edge-middle comparison then gives injectivity of the middle tensor. The endpoint reduction then gives the one-edge three-site statement `thm:peps_edgeBlockedThreeSiteInjective_of_singletons`, and applying this to all edges satisfying $M_e \neq \varnothing$ gives `thm:peps_edgeBlockedThreeSiteInjective_all_of_singletons`. The graph-cardinality variant TNLean.PEPS.EdgeMiddleRegionInjectivityComparison.edgeBlockedThreeSiteInjective˙all˙two˙lt˙card packages the common case $2 < |V|$: since $|M_e| = |V|-2 > 0$ for every edge, the nonempty-middle hypothesis is automatic. Vertex injectivity supplies the two endpoint assertions by TNLean.PEPS.IsVertexInjective.edgeBlockedEndpointTensorMaps˙injective and, for all edges at once, by TNLean.PEPS.IsVertexInjective.edgeBlockedEndpointTensorMaps˙injective˙all, and TNLean.PEPS.IsVertexInjective.edgeBlockedThreeSiteInjective˙of˙middle reduces the remaining work to the middle-block injectivity statement. Its all-edge form is TNLean.PEPS.IsVertexInjective.edgeBlockedThreeSiteInjective˙all˙of˙middle.
   The comparison TNLean.PEPS.EdgeMiddleRegionInjectivityComparison records the conditional passage from finite-region injectivity of $V\setminus\{u,v\}$ to the edge-middle tensor family. Its theorem TNLean.PEPS.EdgeMiddleRegionInjectivityComparison.edgeBloc then combines this comparison with vertex injectivity to obtain the edge-blocked three-site assertion at one edge. The all-edge middle-tensor comparison is TNLean.PEPS.EdgeMiddleRegionInjectivityComparison.edgeMiddleTensorInjective˙all. The all-edge conditional form is TNLean.PEPS.EdgeMiddleRegionInjectivityComparison.edgeBlockedThreeSi The finite induction on exposed middle vertices is separated as TNLean.PEPS.FiniteRegionKernelDescent: for kernel conditions $K_c(S)$ attached to a coefficient family $c = (c_\rho)_\rho$, the implications $K_c(S) \Rightarrow K_c(S \setminus \{j\})$ imply $K_c(M_e) \Rightarrow K_c(\varnothing)$. The edge-middle instance of this coefficient argument is recorded by TNLean.PEPS.EdgeMiddleKernelDescentData: the zero relation $\sum_\rho c_\rho T^\rho_{A,M_e}(\tau) = 0$ for every middle physical index τ gives $K_c(M_e)$, and $K_c(\varnothing)$ gives $c_\rho = 0$ for every ρ. Consequently TNLean.PEPS.EdgeMiddleKernelDescentData.edgeMiddleTens

proves $\mathrm{inj}(T_{A,M_e})$ from such a datum, and TNLean.PEPS.IsVertexInjective.edgeBlockedThreeSiteInjective'of' combines it with the endpoint injectivity assertions. The missing mathematical step is still the contraction theorem used in the paper: contracting vertex-injective tensors over that finite middle region gives an injective blocked tensor. The full transfer is now stated as TNLean.PEPS.IsVertexInjective.edgeBlockedThreeSiteInjective; its proof remains open and is tracked by issue #1366.

2. The two internal steps in Lemma `inj_isomorph` must be formalized for the edge-blocked three-site MPS. The local endpoint part of the virtual-to-physical realization $X \mapsto O_1, O_2$ is formalized by `thm:peps_virtualInsertionPhysicalRealization`: vertex injectivity turns the one-leg virtual matrix action at either endpoint into a physical operator on that endpoint. The endpoint coefficient-level direction is formalized by `thm:peps_edgeInsertedCoeff_endpointPhysicalRealization`: after fixing ordinary edge-boundary data, the left and right physical operators supply the independent inserted-edge index at the corresponding endpoint. On the left endpoint the formal statement uses the transpose of the inserted matrix, because the ordinary boundary supplies the right bond index. The converse recovery $O_1, O_2 \mapsto W$ is stated as TNLean.PEPS.physical'to'virtual'insertion; its proof remains open and is tracked by issue #1370. Its local endpoint recovery component is now formalized by TNLean.PEPS.edgeEndpointLocalVirtualOpOfPhysicalOp'eq'of'projected'realization'eq: once the compressed endpoint actions are known to realize the same bond matrix, the local virtual pullbacks recover that matrix on the shared bond.

3. Equality of PEPS states must be upgraded to equality of edge-inserted coefficients after the three-site reduction. This is the step that produces the matrix-algebra isomorphism on the chosen bond. The formally stated theorem is TNLean.PEPS.isEdgeBlockedInsertionAlgebraIsomorphism; its proof remains open and is tracked by issue #1367.

4. The algebra isomorphism must be converted into an invertible edge gauge, using the same finite-dimensional matrix-algebra argument as the one-dimensional injective theorem. In the blueprint this is the not-yet-formalized theorem `thm:peps_edgeGaugeFromInsertionAlgebraIsomorphism`.

5. After all edge gauges have been absorbed into the second tensor family, producing the modified tensors $\widetilde{B}_v$, the formalization must prove the analog of `eq:inj_equal_edge`: arbitrary insertions of the same matrix on the same edge give equal PEPS states for the first tensor family and the modified second tensor family. In the blueprint the absorption step is the not-yet-formalized theorem `thm:peps_edgeGaugeAbsorption`, and the post-absorption insertion equality target is now named formally by TNLean.PEPS.PostAbsorptionEdgeInsertionEquality. Constructing that

predicate from the absorbed edge gauges remains the not-yet-formalized theorem `thm:peps_postAbsorptionEdgeInsertionEquality`.

6. The generalized two-injective-tensor comparison from `lem:inj_equal_tensors_2` must be formalized. It says that equality of all virtual-bond insertions for two pairs of injective tensors forces $A_1 = \lambda B_1$ and $A_2 = \lambda^{-1} B_2$. In the blueprint this is split into the scalar-reduction substep `thm:peps_twoInjectiveGaugeScalarReduction` and the formally stated comparison theorem TNLean.PEPS.two˙injective˙tensor˙insertion˙comparison. The formal statement is an abstract version with nonempty spectator external boundary spaces; the source lemma is recovered by taking these spaces to be one-point spaces. Its proof remains open and is tracked by issue #1361.

7. The final two-tensor comparison must then be applied by blocking one vertex against its complement. The paper uses this to prove $A_i = \lambda_i \widetilde{B}_i$ at each vertex, and then incorporates the scalars into the local gauges. This specialization is now represented by TNLean.PEPS.one˙vertex˙complement˙comparison. The formal statement uses the abstract two-block notation with nonempty external boundary spaces; the source comparison is obtained from the one-vertex and complement blocks after `eq:inj_equal_edge`.

8. The boundary-insertion argument must remove the separate bond-dimension-identification hypothesis in TNLean.PEPS.fundamentalTheorem˙PEPS˙of˙bondDim. This is the remaining difference between that conditional theorem and the source theorem; it is tracked by issue #874.

4 Source-to-formalization ledger

The following ledger records the present status of the Section 3 proof route. It is meant to prevent the proof from drifting away from the source argument.

- `eq:block_to_mps` edge blocking: `edgeBlockedCoeff_eq_stateCoeff` is locally proved; the injectivity bridge is stated as TNLean.PEPS.IsVertexInjective.edgeBlockedThreeSiteInject with proof tracked by issue #1366. The middle physical index of this three-site object is formalized by TNLean.PEPS.EdgeMiddlePhysicalConfig. The named injectivity statement is TNLean.PEPS.EdgeBlockedThreeSiteInjective, and the endpoint assertions are proved from vertex injectivity, both for a single chosen edge and uniformly over all edges. The comparison from edge-middle region injectivity to edge-middle tensor injectivity is also available uniformly over all edges. The all-edge conditional bridge from middle-region injectivity is TNLean.PEPS.EdgeMiddleRegionInjectivityComparison.edgeBlockedThreeSiteInje

- Identity insertion on the chosen bond: `thm:peps_edgeInsertedCoeff_identity` is formalized by finite diagonal reindexing.

- Lemma `inj_isomorph`, $X \mapsto O_1, O_2$: `thm:peps_virtualInsertionPhysicalRealization` formalizes the endpoint realization, and `thm:peps_edgeInsertedCoeff_endpointPhysicalRealization` formalizes the endpoint coefficient expansion. These are tracked by issues #1369 and #1995.
- Lemma `inj_isomorph`, $O_1, O_2 \mapsto W$: TNLean.PEPS.physical˙to˙virtual˙insertion states the source recovery step; its proof remains open and is tracked by issue #1370. The endpoint projected-recovery component for a common bond matrix is formalized by TNLean.PEPS.edgeEndpointLocalVirtualOpOfPhysicalOp˙eq˙of˙projected˙realiz
- Matrix-insertion algebra isomorphism: TNLean.PEPS.isEdgeBlockedInsertionAlgebraIsomorphism states the edge-blocked matrix-algebra correspondence and its inserted-coefficient equality. Its proof is tracked by issue #1367.
- Edge gauge from the algebra isomorphism: `thm:peps_edgeGaugeFromInsertionAlgebraIsomorphism`, tracked by issue #1368.
- Absorption of edge gauges into B: `thm:peps_edgeGaugeAbsorption`, tracked by issue #1363.
- Equation `eq:inj_equal_edge`: `thm:peps_postAbsorptionEdgeInsertionEquality`, tracked by issue #1364.
- Lemma `inj_equal_tensors_2`, scalar step: `thm:peps_twoInjectiveGaugeScalarReduction`, tracked by issue #1362.
- Lemma `inj_equal_tensors_2`, comparison: TNLean.PEPS.two˙injective˙tensor˙insertion˙comparison states the abstract two-block insertion comparison with nonempty external boundary spaces and reciprocal scalar proportionality. The source lemma is the one-point external-boundary specialization; the proof is tracked by issue #1361.
- One vertex versus its complement: TNLean.PEPS.one˙vertex˙complement˙comparison records the application of the generalized two-block comparison to a selected vertex and its complement, under the abstract nonempty external-boundary formulation. The source proof is tracked by issue #1360.
- Final gauge equivalence and uniqueness: `thm:fundamentalTheorem_PEPS_of_bondDim` and the uniqueness theorem, tracked by issues #780, #842, and the boundary-insertion issue #874.

5 Blueprint diagrams

The blueprint now uses semantic tensor-network commands for the Section 3 moves that should remain visually attached to the proof.

- \TNPEPSEdgeBlockingReduction depicts the chosen edge and the passage to the three-site chain. Its endpoint labels follow eq:block_to_mps: the two circled endpoints are A'_1 and A'_2, while the boxed complement is A'_3.
- \TNPEPSEdgeInsertedCoeff depicts the single edge-blocked contraction with one virtual matrix X inserted on the chosen bond. This is the diagram attached to the definition of the inserted-edge coefficient, before comparing two tensor networks.
- \TNPEPSThreeSiteInsertionComparison depicts the comparison of a virtual insertion in the two three-site chains, as in the statement of lem:inj_isomorph.
- \TNPEPSInsertionPhysicalRealization depicts the realization of a virtual insertion as a physical operation on either neighboring site.
- \TNPEPSPhysicalToVirtualInsertion depicts the converse step in which two neighboring physical operations determine a virtual operation on the shared bond.
- \TNPEPSEdgeInsertionEquality depicts the full PEPS comparison after the edge gauges have been absorbed into the second tensor family, matching the five-vertex graph and the distinguished $(2, 5)$-edge insertion in eq:inj_equal_edge.
- \TNPEPSEdgeGaugeAbsorption depicts the local absorption of incident oriented edge gauges into a B-tensor, producing $\widetilde{B}_v$.
- \TNPEPSTwoInjectiveTensorInsertionComparison depicts the generalized two-injective-tensor comparison with an arbitrary matrix inserted on a shared virtual bond.
- \TNPEPSTwoInjectiveGaugeScalarReduction depicts the residual virtual operators Z, U, W that appear after applying an inverse for one injective block in the proof of lem:inj_equal_tensors_2.
- \TNPEPSOneVertexComplementComparison depicts the final one-vertex-versus-complement comparison used to prove $A_v \propto \widetilde{B}_v$. It follows the source paper's displayed comparison by drawing the one-vertex enlargement and the smaller injective region as tensor-network tensors, rather than as an auxiliary formal block.

These diagrams deliberately keep the local dot convention used in the blueprint, but they preserve the distinctions that matter in the paper: tensor sites, blocked regions, virtual matrix insertions, and physical operations are not drawn as the same object.

6 Conclusion

The current formalization has reached the edge-blocked coefficient and has begun the matrix-insertion layer needed for the Section 3 proof. It has not yet proved the three-site injectivity transfer, the insertion comparison, the resulting edge gauge construction, or the final one-vertex-versus-complement comparison. Those are the remaining mathematical steps required to turn the current PEPS definitions into a proof of the injective PEPS Fundamental Theorem faithful to [MGRPG$^+$18, Section 3].

References

[MGRPG$^+$18] András Molnár, José Garre-Rubio, David Pérez-García, Norbert Schuch, and J. Ignacio Cirac. Normal projected entangled pair states generating the same state. *New Journal of Physics*, 20(11):113017, 2018.